

The background features a light blue gradient. In the top left, there is a pinkish-orange curved shape with several dark green dots of varying sizes. In the top right, there is a light green curved shape with a dark green leafy branch extending from it. On the left side, there are several concentric green circles.

Backend Development

Project 1

Industrial Training Kit

Batch: 2026 | Powered by DecodeLabs

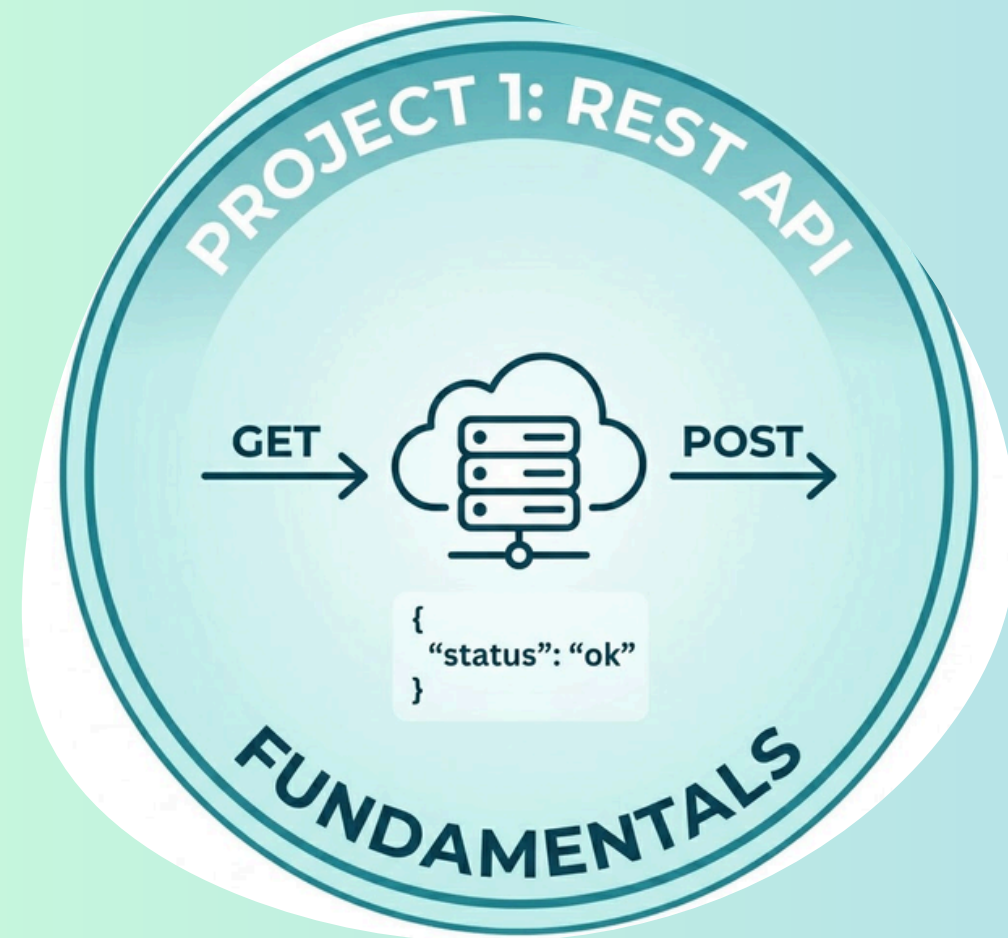
WELCOME TO THE TEAM!

Step into the role of a Backend Developer at DecodeLabs. Project 1 is your essential start: REST API Fundamentals. This track isn't about "just writing code"—it's about Stateless Communication.

Before you build complex databases, you must master the art of creating local servers, defining GET and POST routes, and serving structured JSON data. By completing this milestone, you are proving you can establish the core communication layer of the web through pure routing logic. Let's design a digital foundation that serves data with absolute clarity.



Project 1: REST API Fundamentals



Goal: Build a stateless web server that serves data.

Key Requirements:

Create a local server
(Node.js/Python).

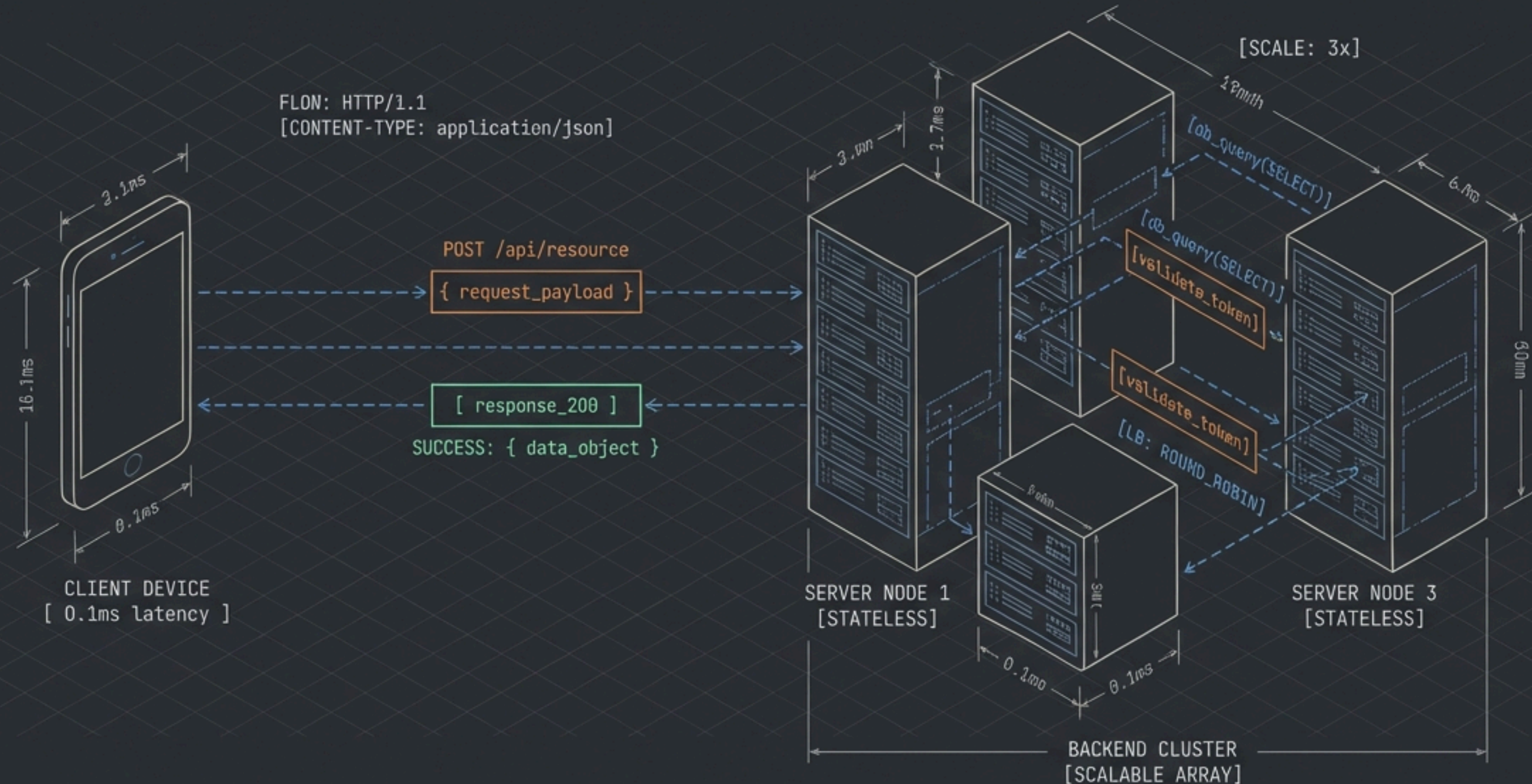
Define GET and POST routes.

Return structured JSON data.

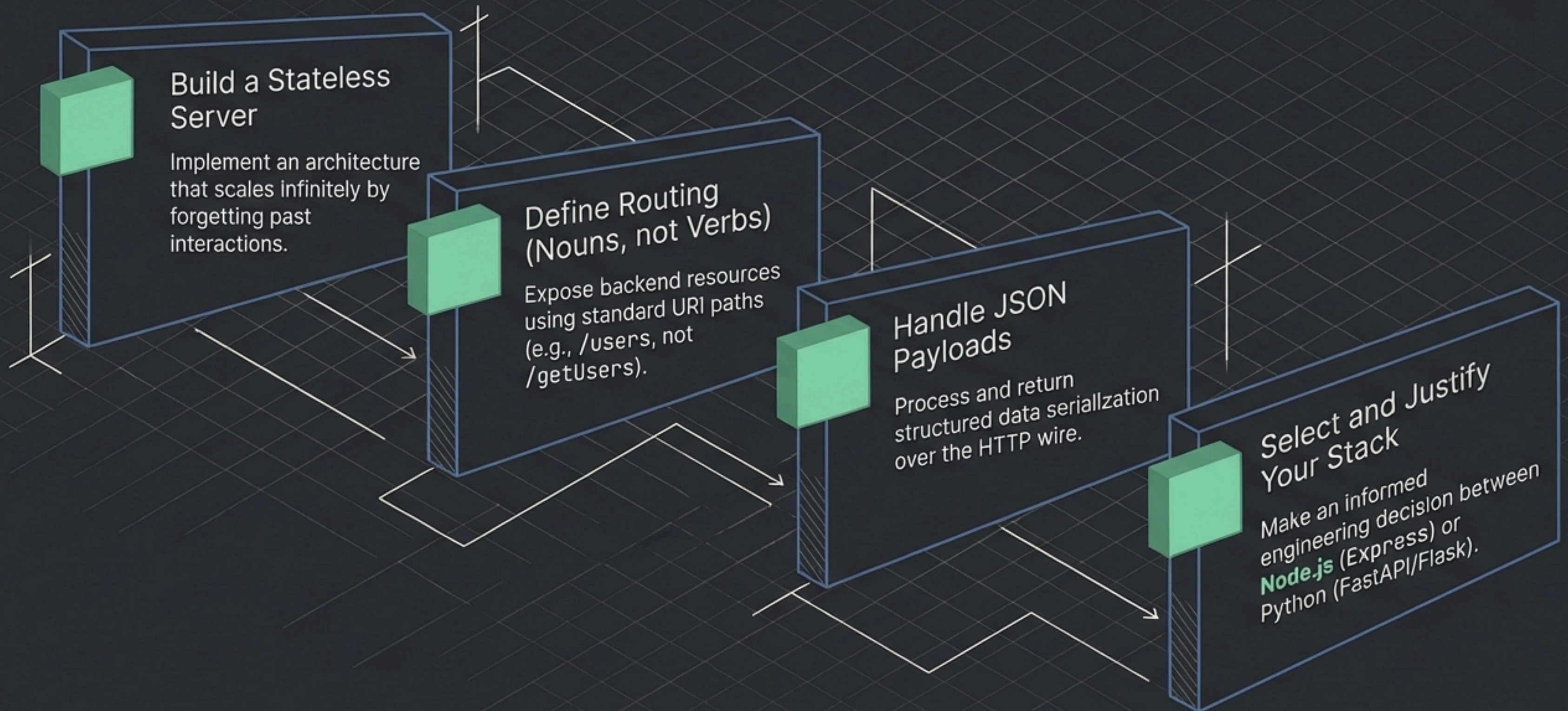
Key Skills: HTTP methods,
routing logic, JSON formatting.

Project 1: REST API Fundamentals

Your blueprint for building stateless, scalable backend systems.



Welcome to the Backend

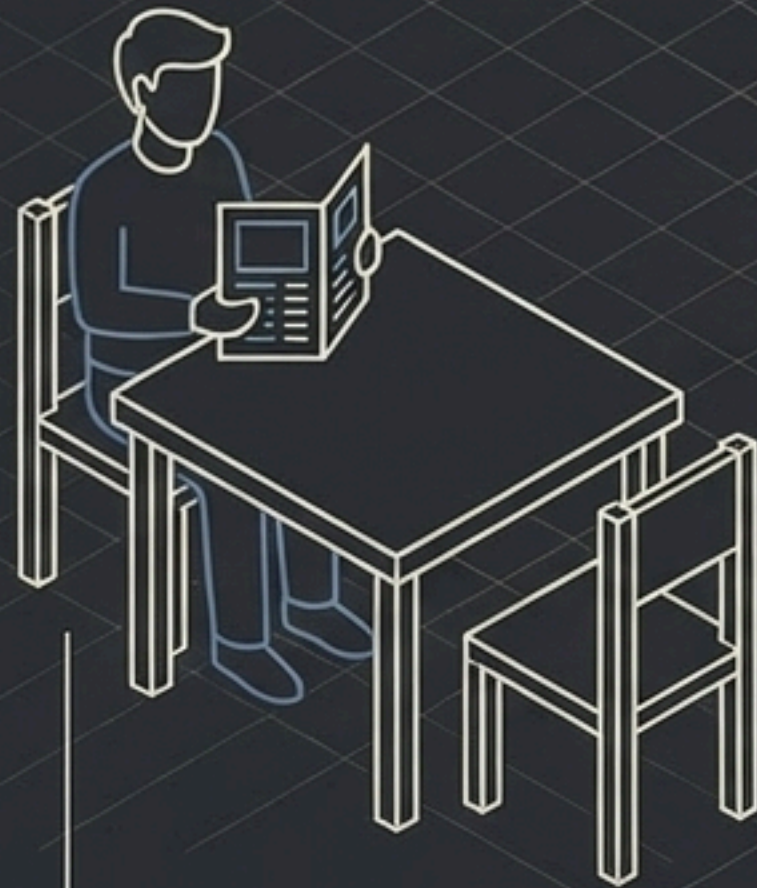


The API is the Digital Waiter

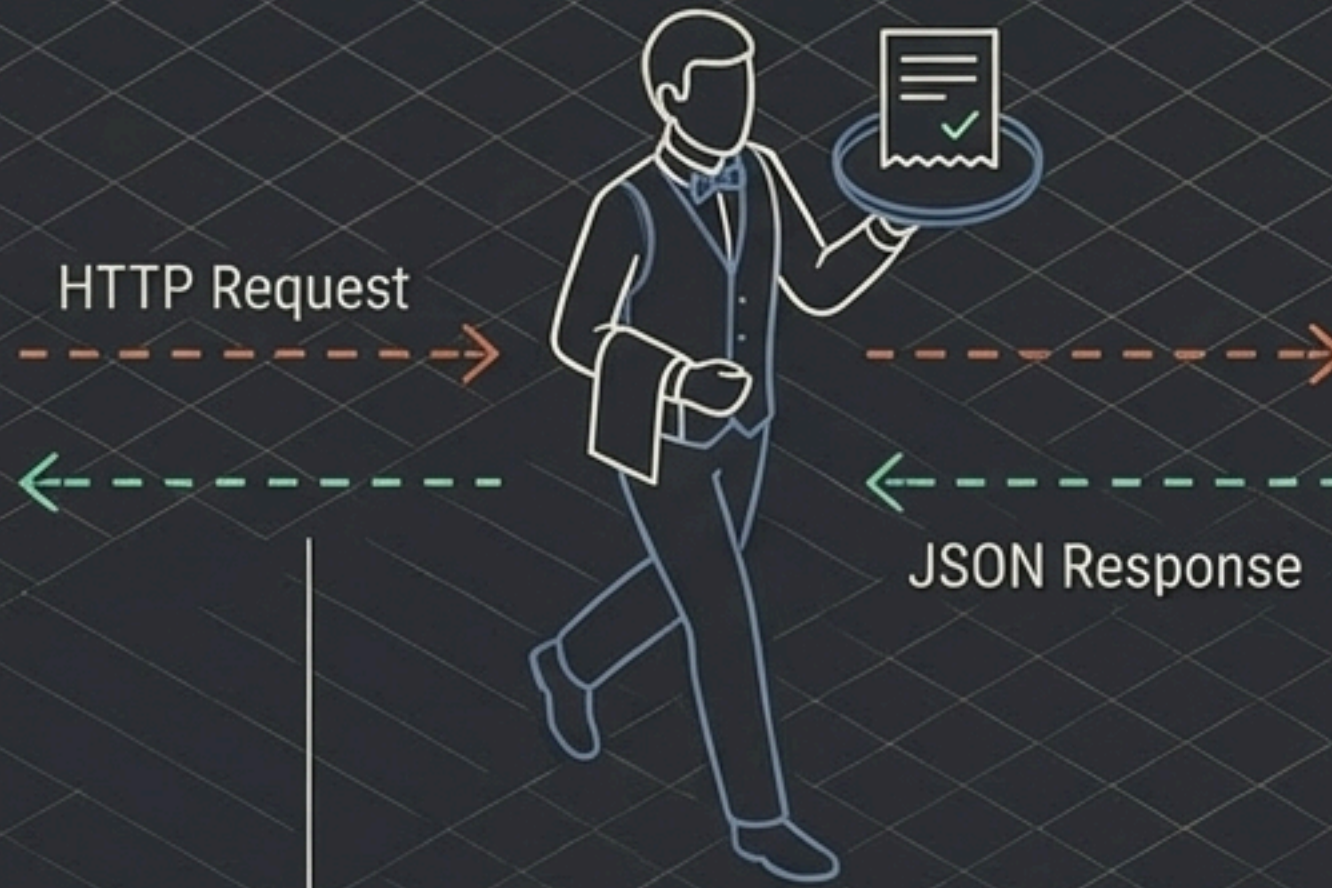
The Client
(Frontend / Mobile App)

The API
(The Messenger)

The Server & Database
(The Kitchen)



Creates the
HTTP Order

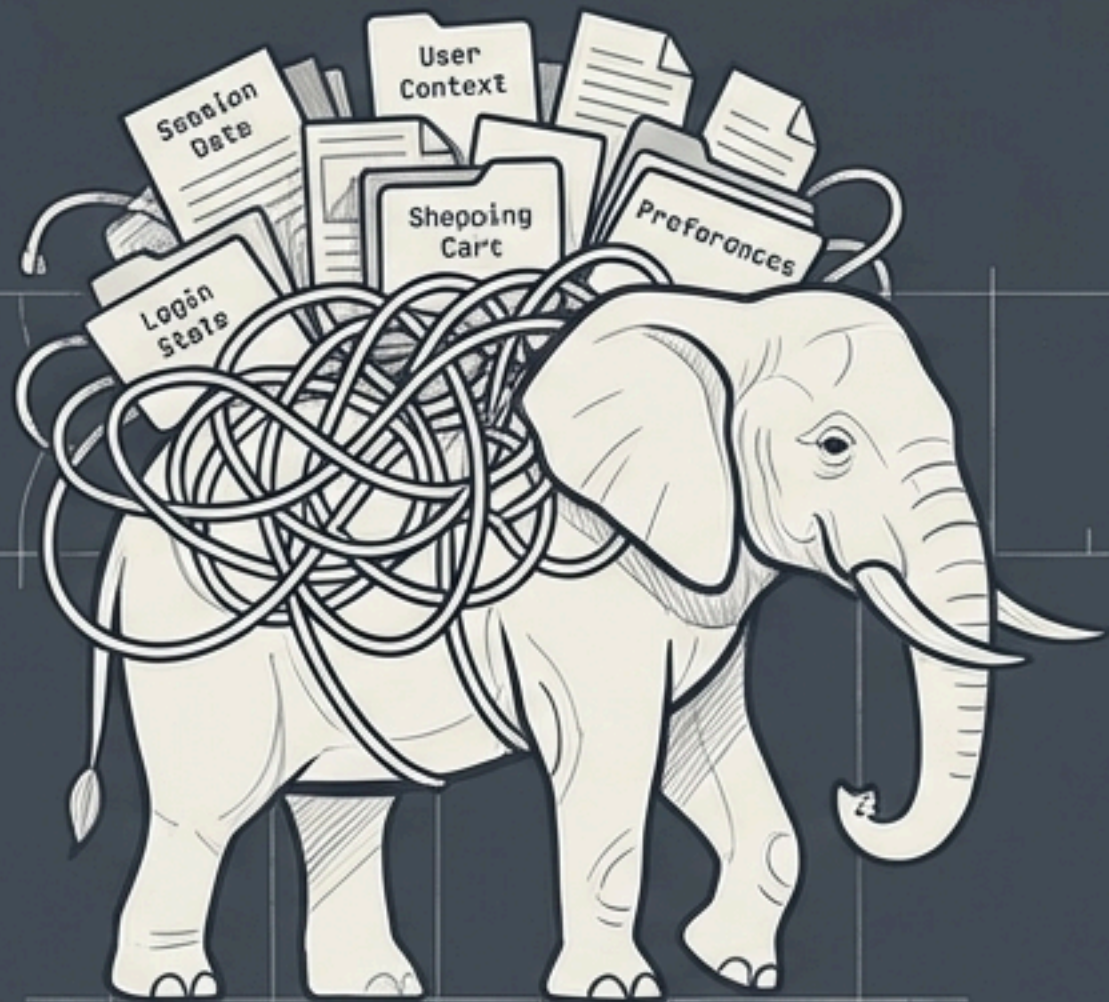


Translates request and
returns the JSON Response



Processes order and
interacts with data

The Golden Rule: The Goldfish Server



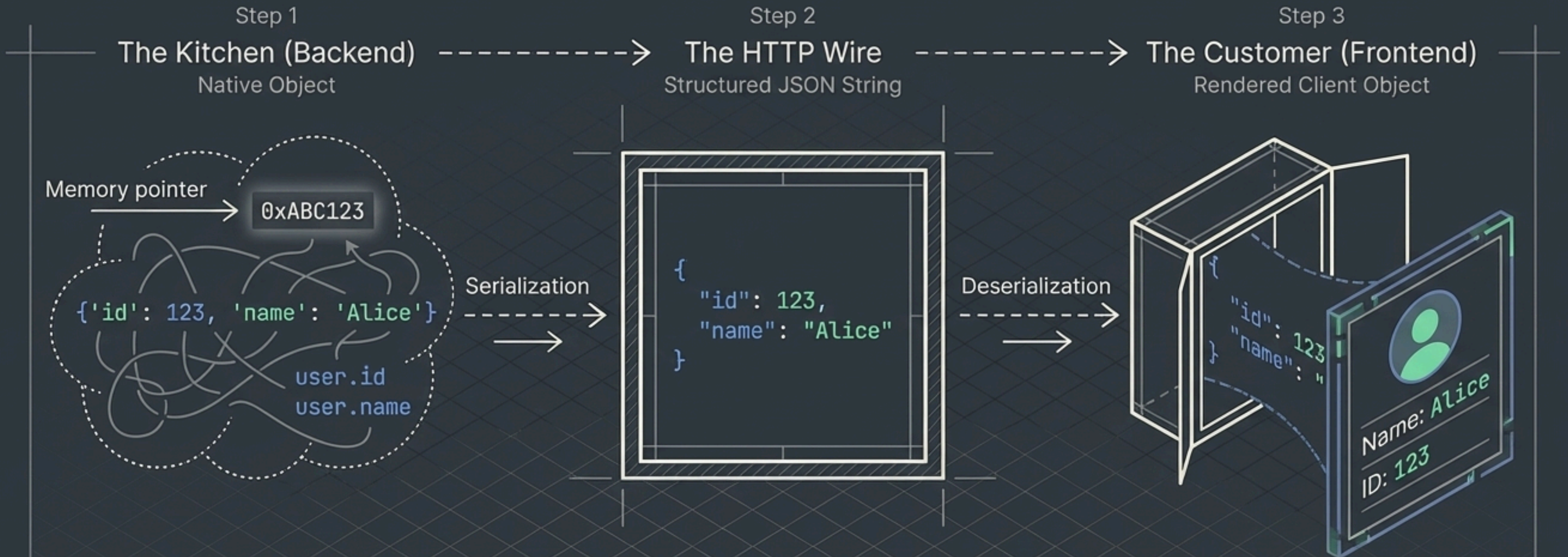
Stateful servers remember you. This breaks horizontal scaling because the server becomes heavy with session context.



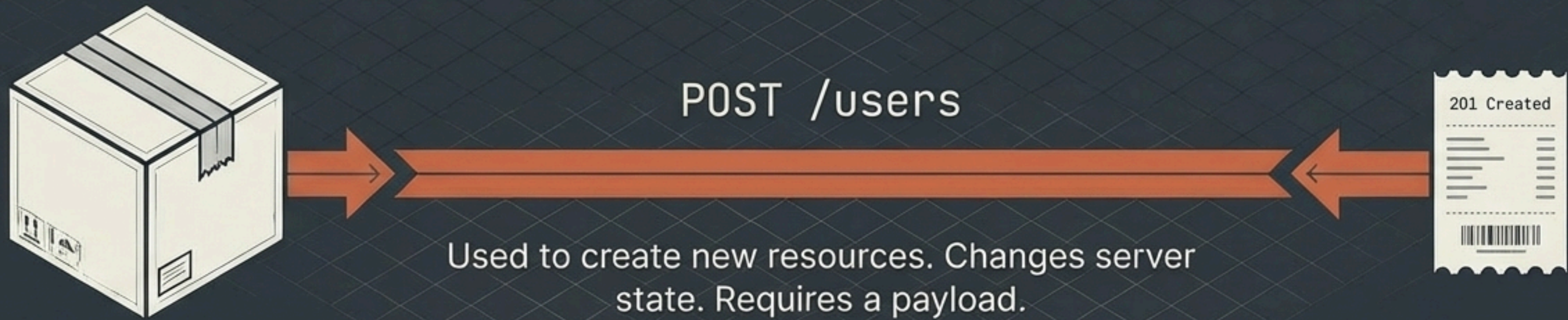
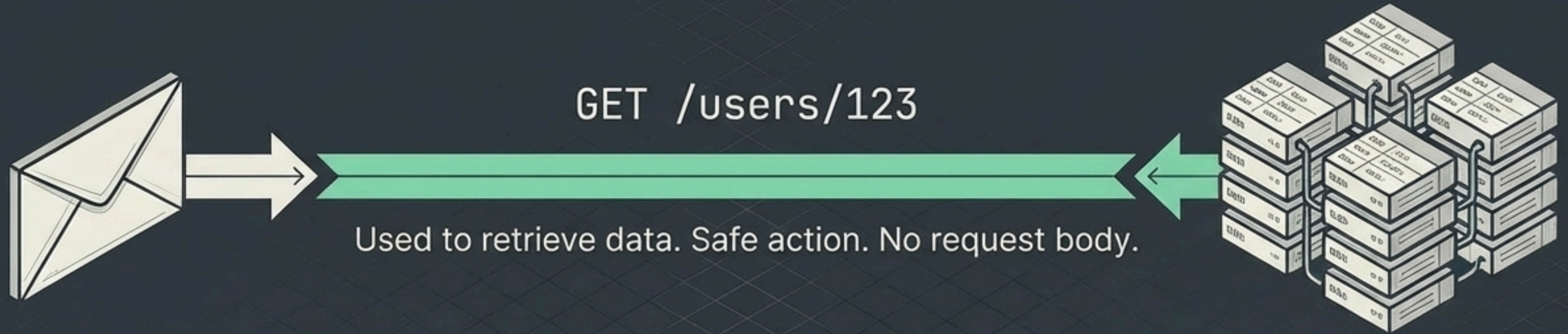
REST servers are goldfish. They forget you the moment the transaction ends.

Key Insight: Every single HTTP request must be fully self-contained. The client must bring its Auth token and full context every time, allowing load balancers to route requests to any server effortlessly.

The Universal Transport Language: JSON

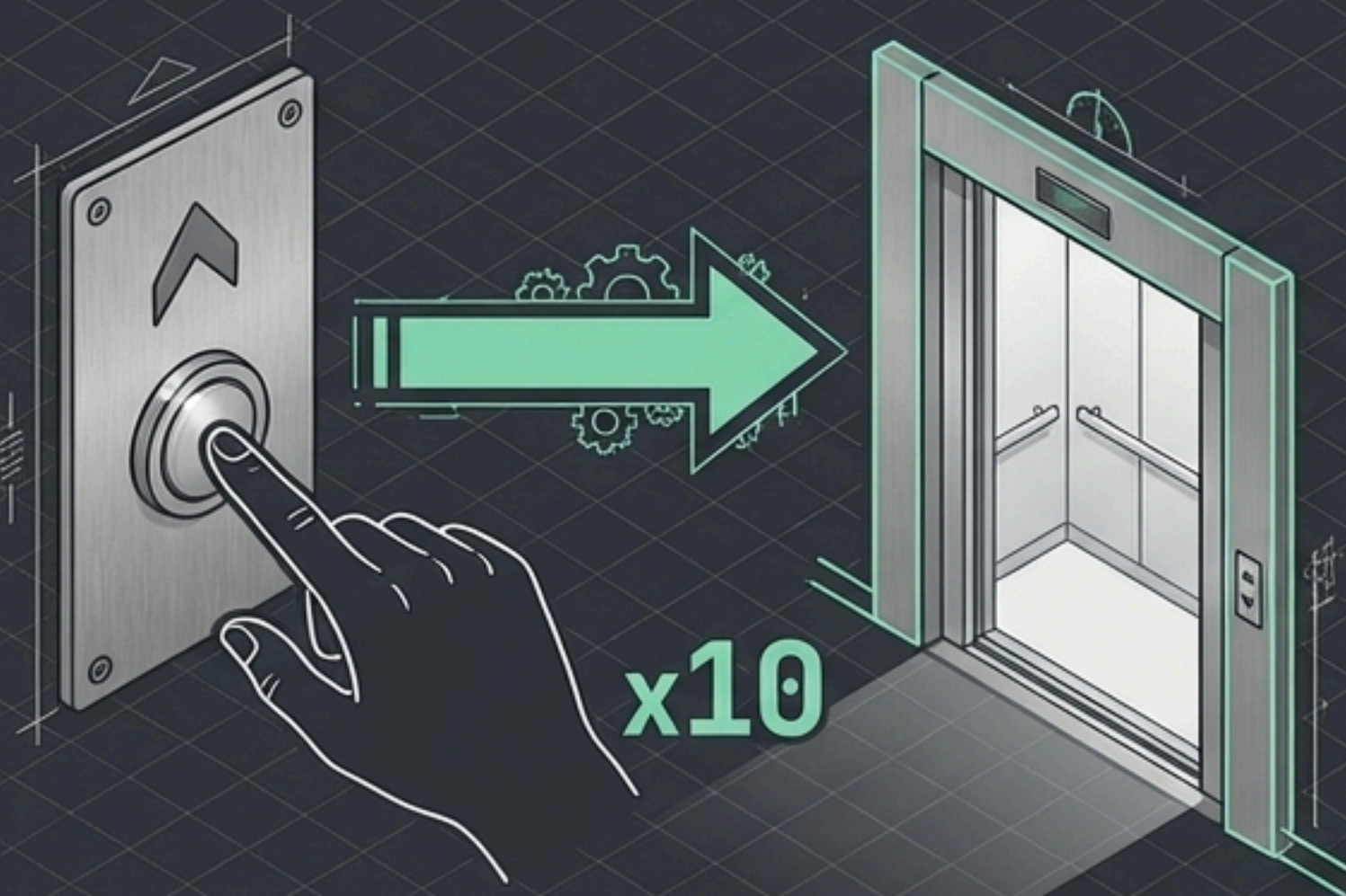


The Core HTTP Verbs



Safe to Retry: Understanding Idempotency

Idempotent: GET, PUT, DELETE



Multiple identical requests yield the **exact same server state** as a single request. **Safe to automatically retry** on network failure.

Non-Idempotent: POST



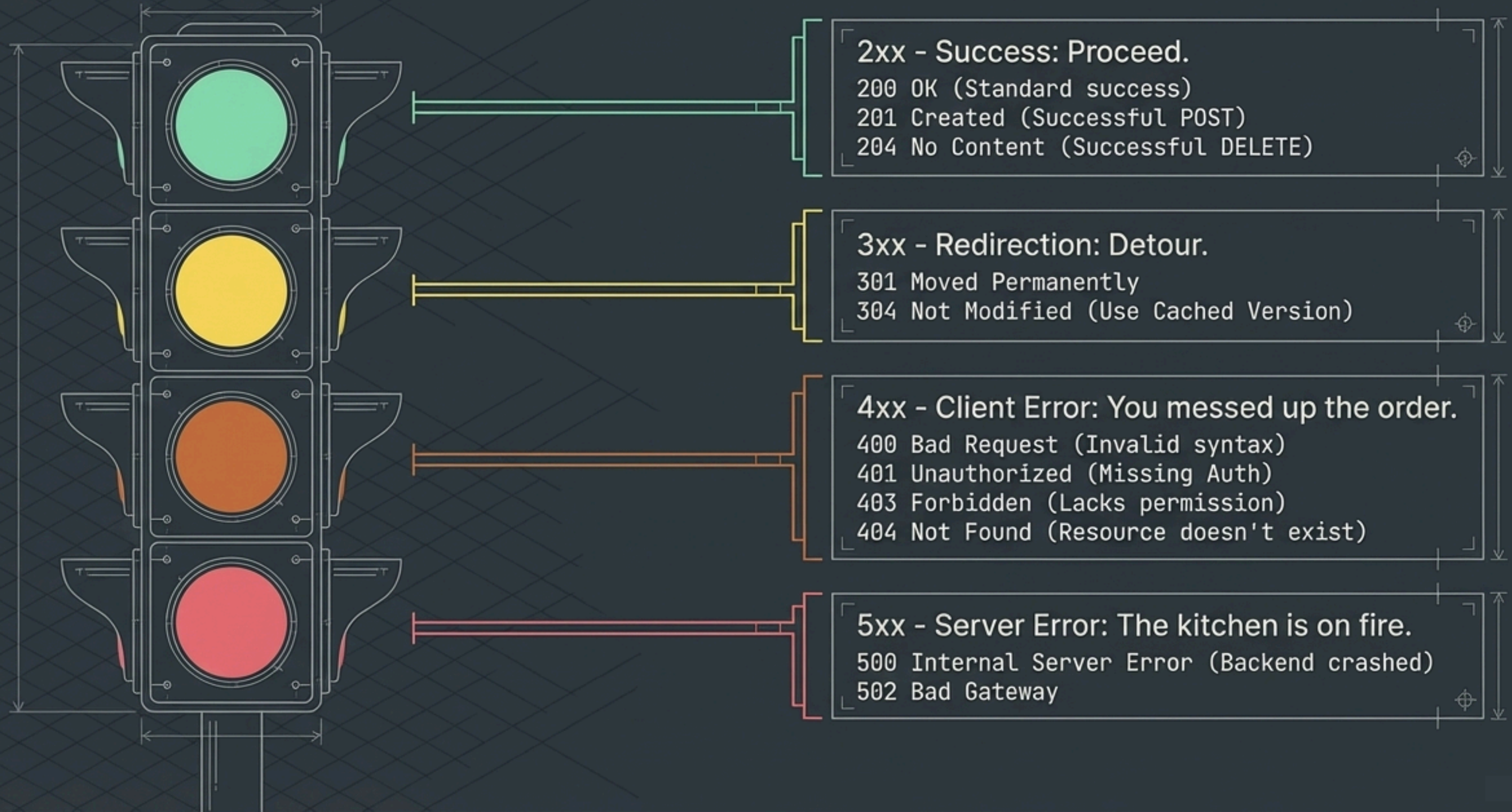
Creates a new resource every time. Requires careful handling (e.g., Idempotency keys) to **prevent duplicate side-effects.**

The HTTP Method Diagnostic Matrix

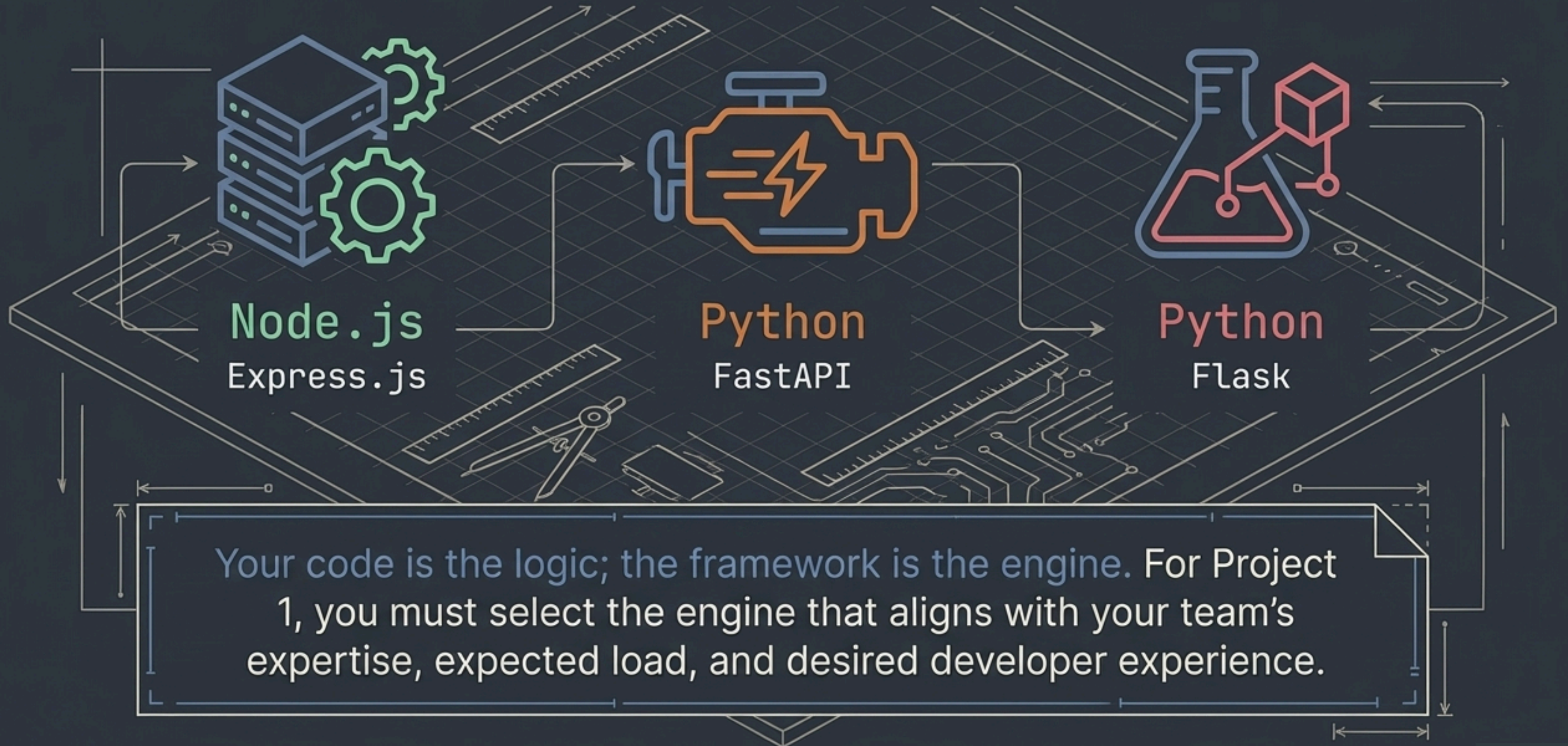
Method	Action	Data Location	Safe	Idempotent
GET	Read	URL / Query	Yes	Yes
POST	Create	JSON Body	No	No
PUT	Full Update / Replace	JSON Body	No	Yes
DELETE	Remove	URL	No	Yes

"Safe" means it does not alter database state.
"Idempotent" means repeating the request yields the same final state.

Status Codes: The Server's Traffic Light

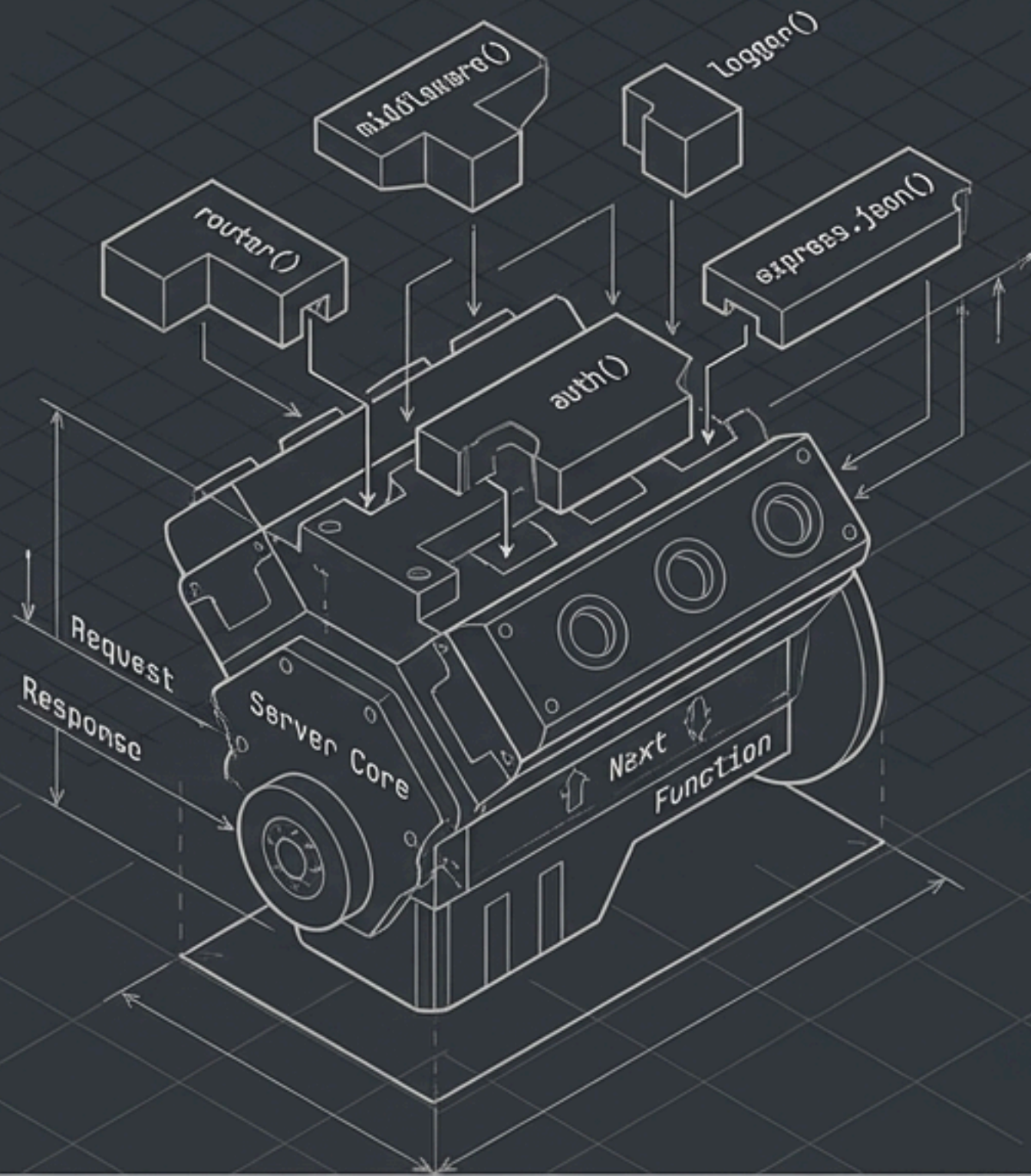


Choosing Your Backend Architecture



Express.js: The Unopinionated Workhorse

Language: JavaScript / Node.js



The Pros

- Unmatched flexibility and lack of strict enforcement.
- Massive npm ecosystem and community support.
- Excellent at I/O-heavy operations (real-time chat, dashboards) utilizing Node's non-blocking event loop.

The Trade-offs

- Requires manual setup for nearly everything.
- Explicitly requires developers to configure JSON parsing via `express.json()` middleware.
- Data validation is completely manual unless third-party libraries are integrated.

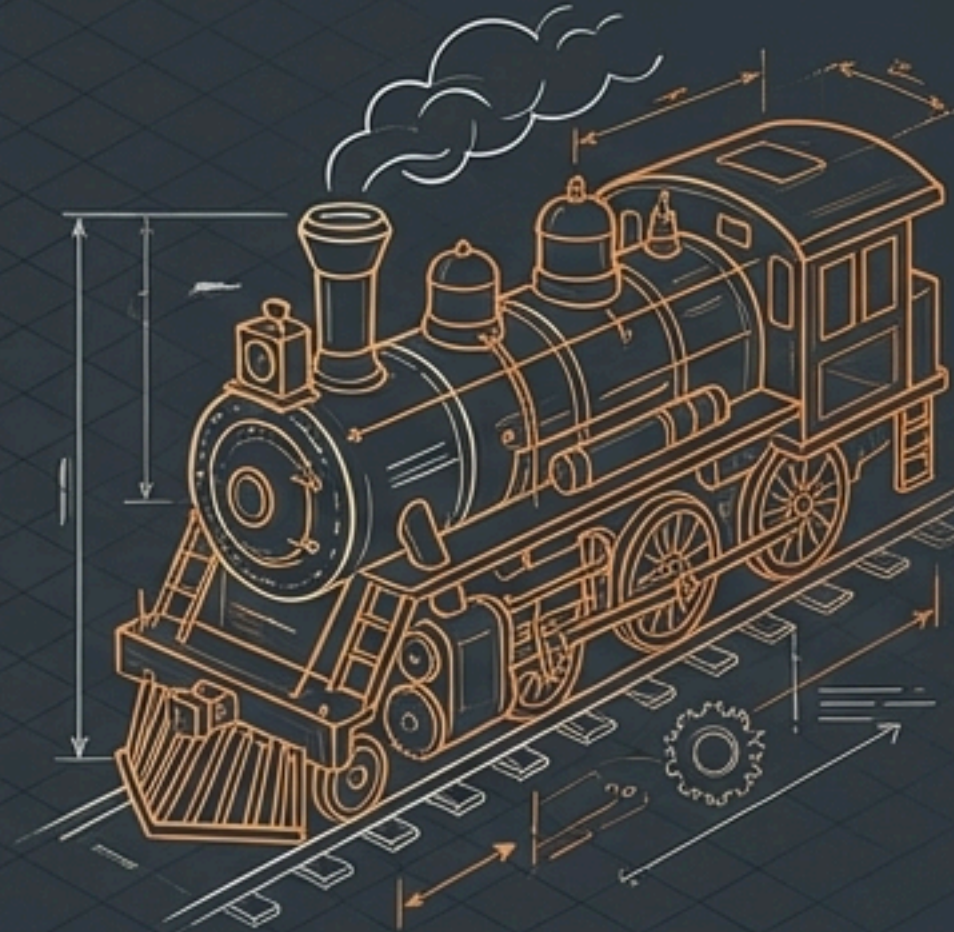
FastAPI vs. Flask: The Python Paradigms

FastAPI: The Modern Speedster



- Built on ASGI for native `async/await` support.
- Blistering speeds (15,000–20,000 requests/sec).
- Automatic JSON validation via Pydantic.
- Auto-generates OpenAPI / Swagger documentation out of the box.

Flask: The Battle-Tested Classic



- Released in 2010. Synchronous WSGI request cycle.
- Stable speeds (2,000–3,000 requests/sec).
- Bare-bones core providing total architectural control.
- Requires manual data validation and manual API documentation.

The Tech Stack Showdown

Dimension	Express.js	FastAPI	Flask
Concurrency Model	Async Event Loop	Native ASGI Async	Synchronous WSGI
Raw Speed	High	15k - 20k RPS	2k - 3k RPS
Data Validation	Manual / External	Automatic (Pydantic)	Manual / External
API Documentation	Manual	Auto-generated (OpenAPI)	Manual
Best For...	Real-time JS apps & dashboards	High-performance, Data/ML APIs	Prototyping & legacy Python integrations

Synthesis: Anatomy of an API Route

The HTTP Method. Dictates a 'Create' action on a plural noun resource.

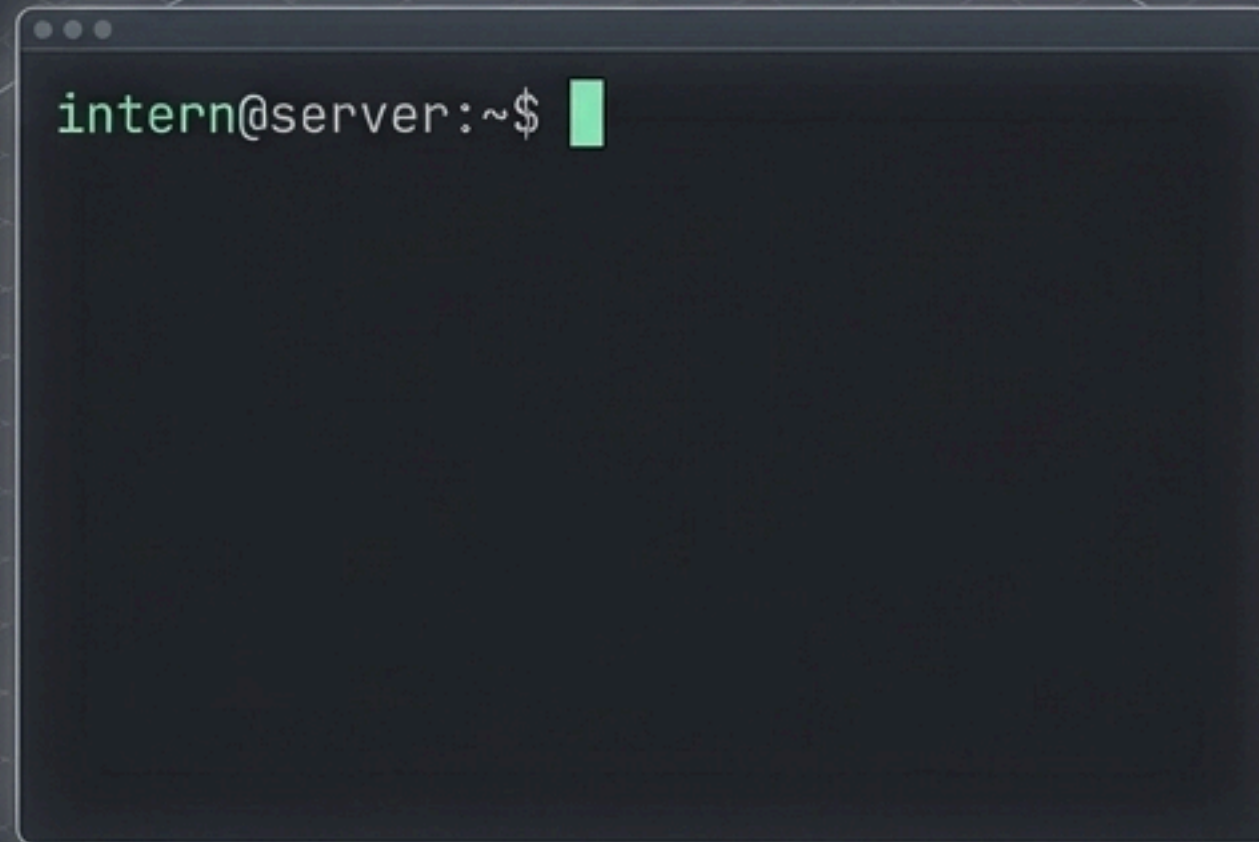
The Traffic Light. Explicitly returning '201 Created' upon success.

```
@app.post("/users", status_code=201)
def create_user(user: User):
    return user
```

JSON Deserialization. Pydantic automatically validates the incoming HTTP body against our contract.

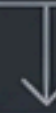
Statelessness. No session variables used. The route processes the request entirely based on the provided JSON payload.

Execution: Project 1 Checklist



Architecture

Select **Express**, **FastAPI**, or **Flask** and configure your base server.



Resource Design

Map out your endpoints using plural nouns (e.g., `/products`, `/orders`).



Implementation

Write your **GET** (Read) and **POST** (Create) routes, ensuring proper JSON parsing (e.g., `express.json()`).



The Contract

Return accurate HTTP status codes (200, 201, 400, 404) and validate your inputs.

The backend is yours to build. Execute the blueprint.

CONCLUSION

The absolute best way to master Backend Development is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like setting up custom error messages or organizing your routes into separate modules—and treat every "500 Internal Server Error" as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your server-side proficiency to future employers. Your journey to becoming a professional developer begins right here, right now, with the very first endpoint you create today.



THANK YOU

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